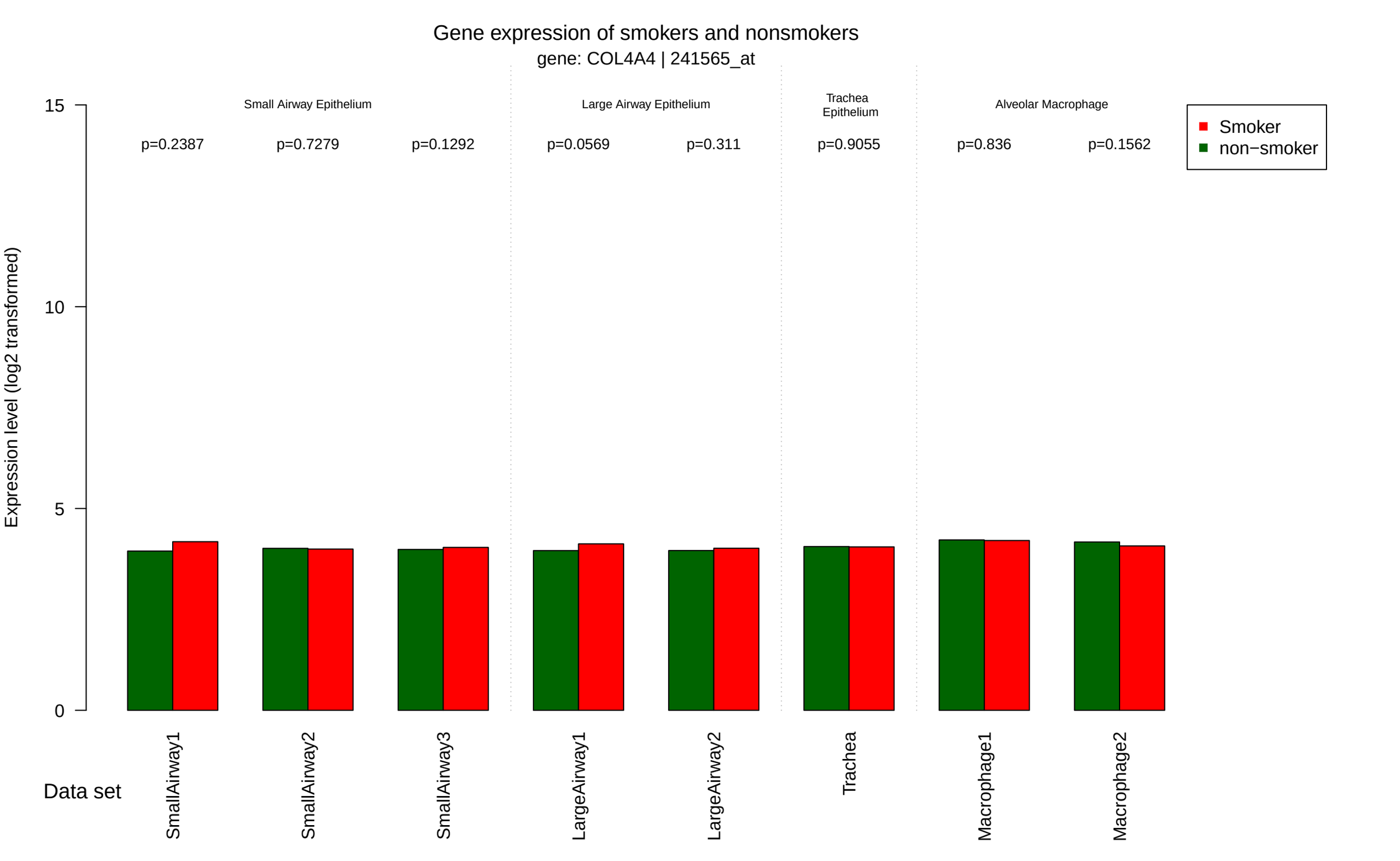
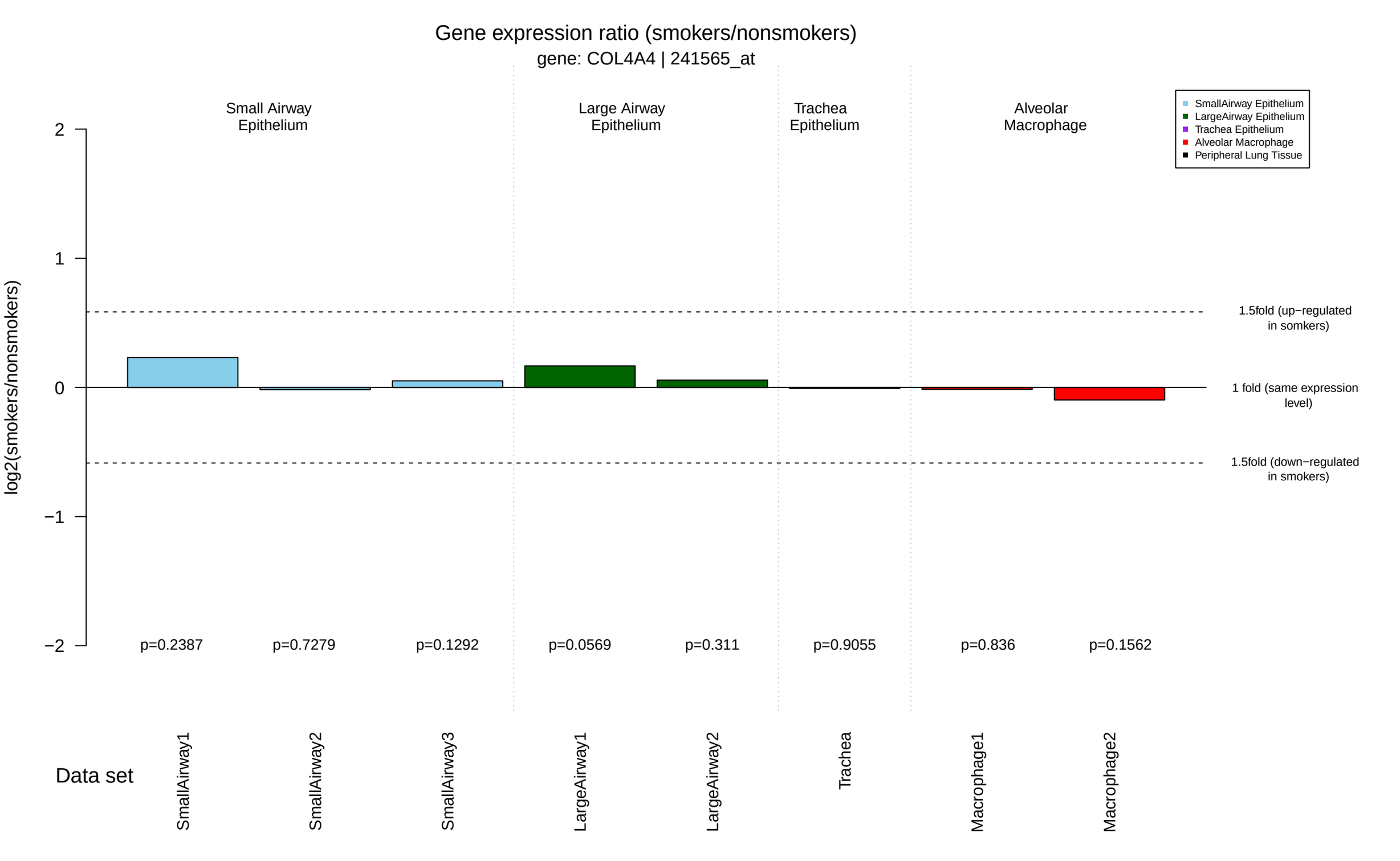
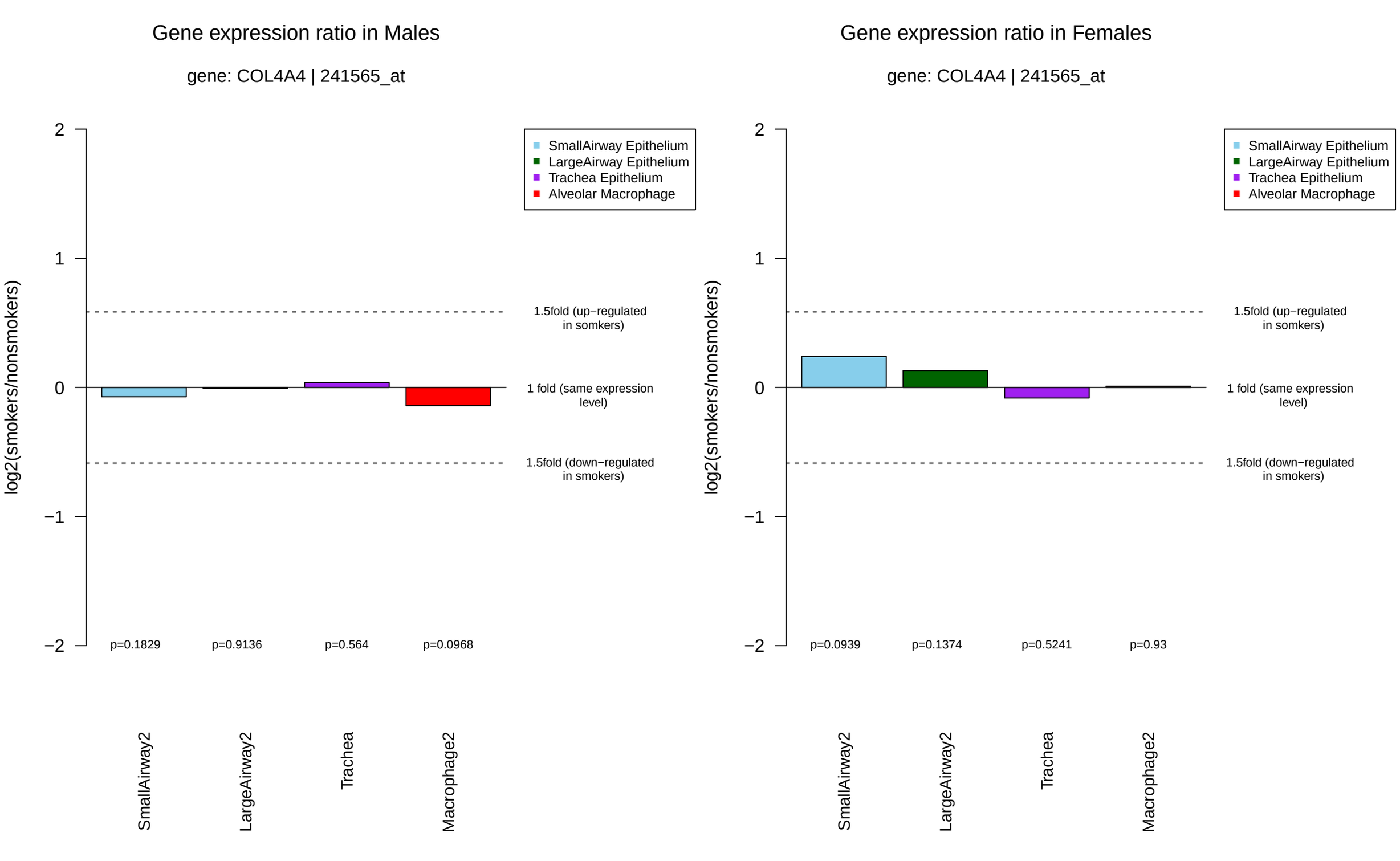
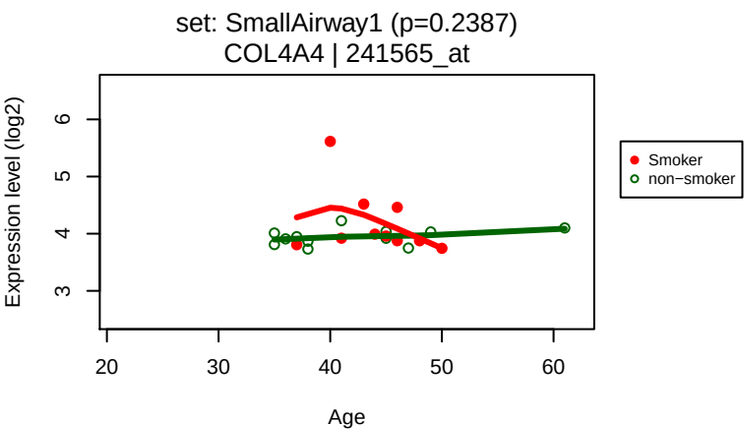
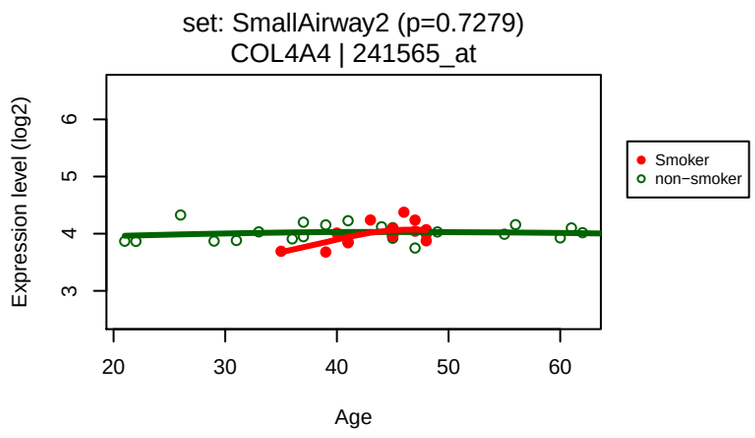
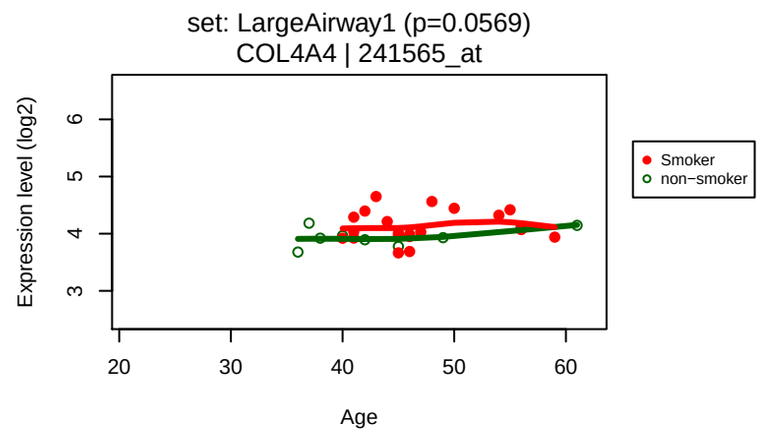

Gene expression of smokers and nonsmokers
gene: COL4A4 | 241565_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2387
p=0.7279
p=0.1292
p=0.0569
p=0.311
p=0.9055
p=0.836
p=0.1562
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

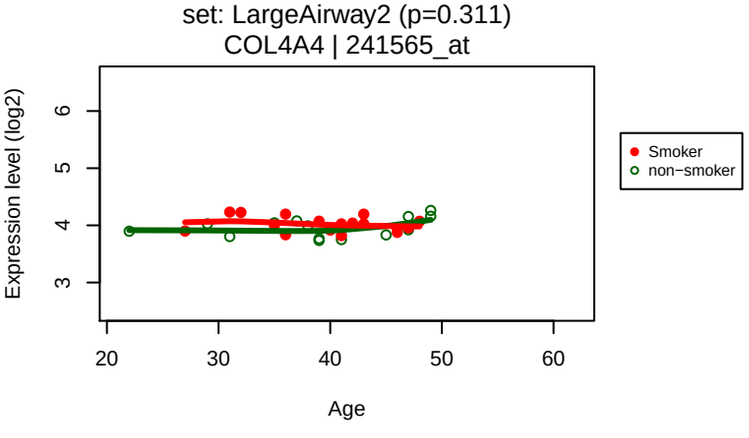

Gene expression ratio (smokers/nonsmokers)
gene: COL4A4 | 241565_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2387
p=0.7279
p=0.1292
p=0.0569
p=0.311
p=0.9055
p=0.836
p=0.1562
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

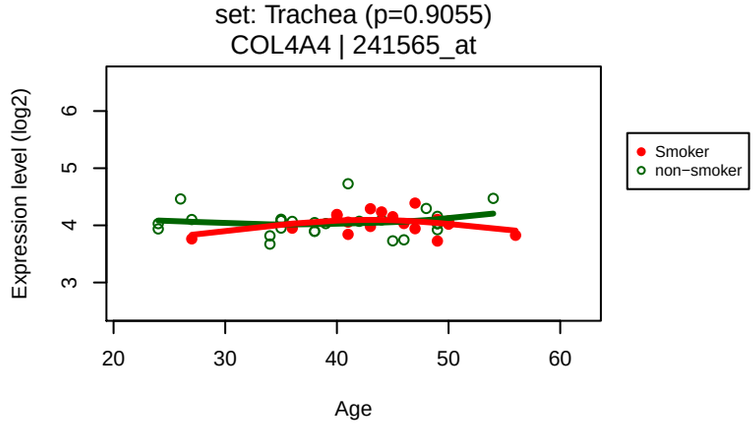

Gene expression ratio in Males
gene: COL4A4 | 241565_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1829
p=0.9136
p=0.564
p=0.0968
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A4 | 241565_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0939
p=0.1374
p=0.5241
p=0.93
SmallAirway2
LargeAirway2
Trachea
Macrophage2

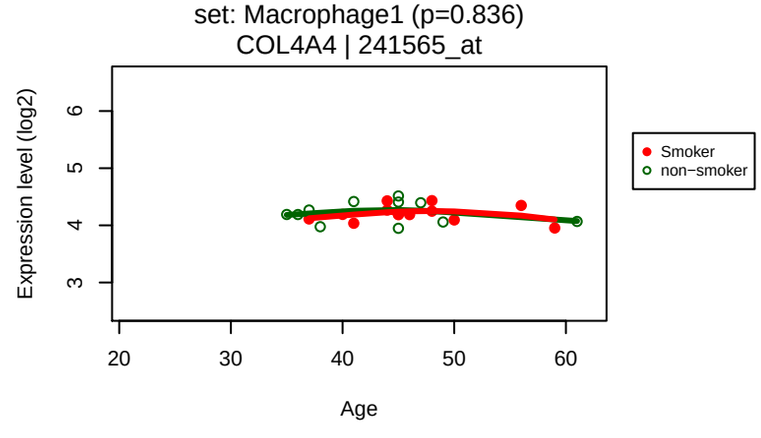

set: SmallAirway1 (p=0.2387)
COL4A4 | 241565_at
Expression level (log2)
Age
Smoker
non−smoker

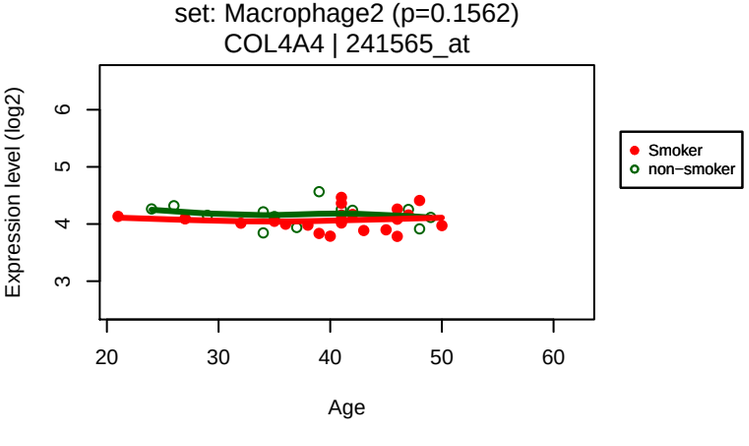

set: SmallAirway2 (p=0.7279)
COL4A4 | 241565_at
Expression level (log2)
Age
Smoker
non−smoker

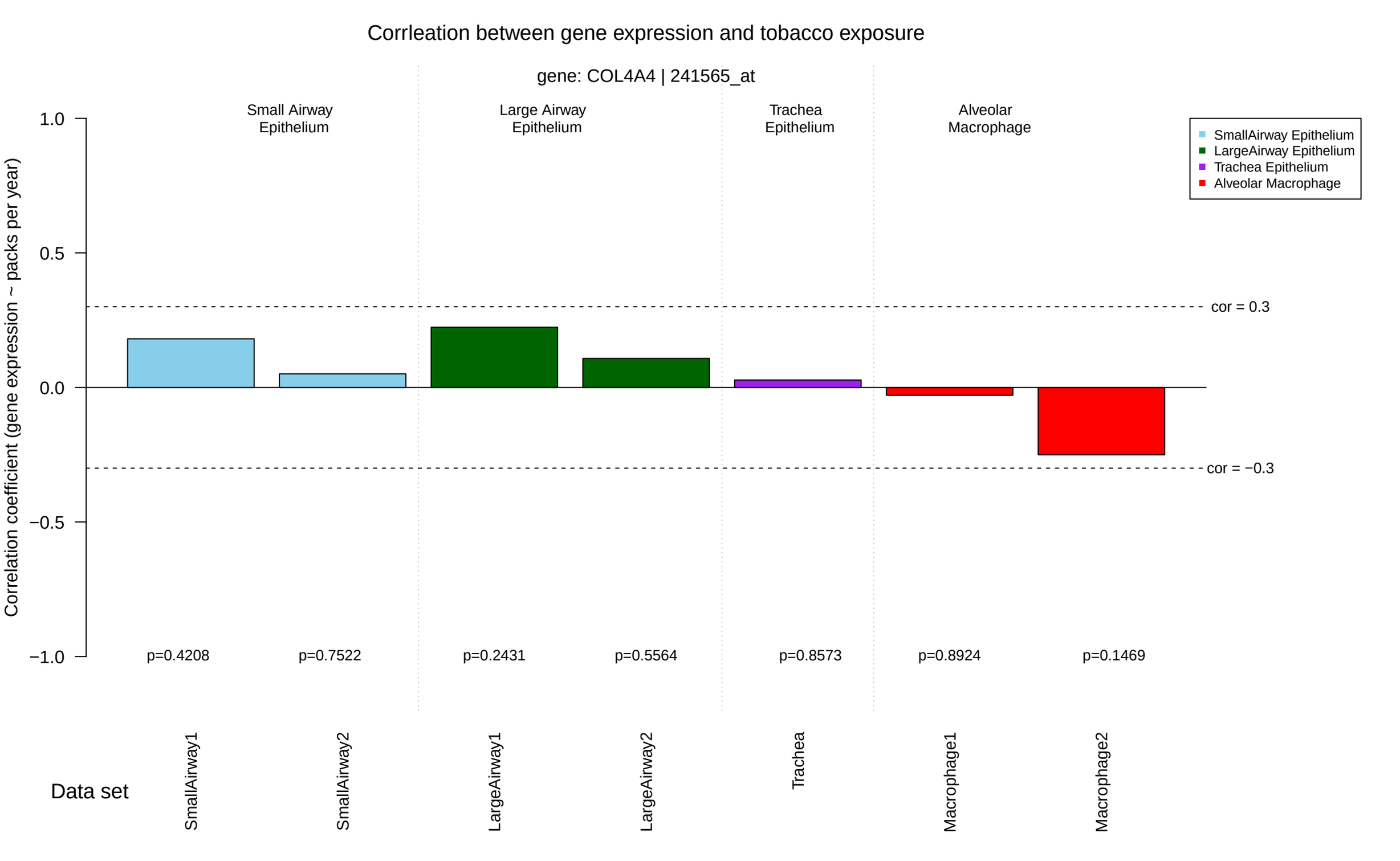

set: LargeAirway1 (p=0.0569)
COL4A4 | 241565_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.311)
COL4A4 | 241565_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9055)
COL4A4 | 241565_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.836)
COL4A4 | 241565_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1562)
COL4A4 | 241565_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A4 | 241565_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4208
p=0.7522
p=0.2431
p=0.5564
p=0.8573
p=0.8924
p=0.1469
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2